GARMIN INTERNATIONAL

Garmin Connect Developer Program
Courses API

Version 2.0

CONFIDENTIAL

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1. Revision History

Version	Date	Revisions
1.0	12/01/2020	Initial revision
1.1	02/02/2020	Updated Course Point Types
2.0	10/15/2026	Garmin Connect Developer Program updated

2. Getting Started

2.1. Purpose of the API

The Garmin Connect Courses API is the underlying mechanism by which users can opt to import courses from third-party platforms into their Garmin Connect account, making it easy to manage this type of information in a centralized location.

2.2. API Rate Limiting or Excessive Usage

To manage capacity and ensure system stability, Garmin Connect Courses API implementations may be subject to rate limiting. If any of the following limits are problematic for your implementation, please contact connect-support@developer.garmin.com to discuss alternatives.

Please plan the implementation with the following limitations in mind:

Starter Level Rate Limits

- 100 API call requests per app per minute - a rolling 60 second window summing the Oauth requests and API calls.
- 200 API call requests per user per day - a rolling 24-hour window excluding Oauth requests.

Enterprise Level Rate limits

- 3000 API call requests per app per minute - a rolling 60 second window summing the Oauth requests and API calls.
- 3000 API call requests per user per day - a rolling 24-hour window excluding Oauth requests.

If one or both of the above limits are exceeded by a app or a specific user, the subsequent API call request attempts will receive an HTTP Status Code 429 (too many requests). The call or calls in question will need to be attempted again later.

3. Courses API

3.1 Field Definitions

Course	Data Type	Description
courseId	Long	a unique identifier for the Courses. This field is not necessary for create operations and will be set automatically.
elapsedSeconds	Double	number of elapsed seconds.

CourseDetails	Data Type	Description
courseName	String	the name of the course, this cannot be empty
description	String	the description of the course
distance	Double	total distance of the course in meters, this cannot be null
elevationGain	Double	total elevation gain of the course in meters and cannot be null
elevationLoss	Double	total elevation loss of the course in meters and cannot be null
geoPoints	List<GeoPoint>	a list of geo-points that constitutes the courses and cannot be empty
activityType	String	activity type of the course, valid Values: RUNNING, HIKING, OTHER, MOUNTAIN_BIKING, TRAIL_RUNNING, ROAD_CYCLING, GRAVEL_CYCLING
speedMetersPerSecond	Double	speed of the total course in meters per second
coordinateSystem	String	valid co-ordinate system values: WGS84, GCJ02, BD09

GeoPoint	Data Type	Description
latitude	Double	latitude for the geo-point
longitude	Double	longitude for the geo-point
elevation	Double	elevation for the geo-point
information	CoursePoint	coursePoint is informational point

CoursePoint	Data Type	Description
coursePointType	String	valid CoursePointType values: GENERIC, SUMMIT, VALLEY, WATER, FOOD, DANGER, FIRST_AID, HORS_CATEGORIE, FOURTH_CATEGORY, THIRD_CATEGORY, SECOND_CATEGORY, FIRST_CATEGORY, SPRINT, SEGMENT_START, SEGMENT_END,SHAPING, CAMPSITE, AID_STATION, REST_AREA, GENERAL_DISTANCE, SERVICE, ENERGY_GEL, SPORTS_DRINK, MILE_MARKER, CHECKPOINT, SHELTER, MEETING_SPOT, OVERLOOK, TOILET, SHOWER, GEAR, SHARP_CURVE, STEEP_INCLINE, TUNNEL, BRIDGE, OBSTACLE, CROSSING, STORE, TRANSITION, NAVAID, TRANSPORT, ALERT, INFO
segmentUuid	String	uniqueId for identifying segment, applicable only when CoursePointType is SEGMENT_START, SEGMENT_END

3.2 Example JSON


```
{
      "courseId": 30626618,
      "courseName": "olathe gravel cycling",
      "distance": 8561.08,
      "elevationGain": 115.27,
      "elevationLoss": 4.44,
      "geoPoints": [{
            "latitude": 46.425274,
            "longitude": 11.685595,
            "elevation": 1300.0
        },
        {
            "latitude": 46.426752,
            "longitude": 11.684781,
            "elevation": 1300.0
        },
        {
            "latitude": 46.429178,
            "longitude": 11.68578,
            "elevation": 1300.4
        },
        {
            "latitude": 46.430211,
            "longitude": 11.685834,
            "elevation": 1304.2
        },
        {
            "latitude": 46.430463,
            "longitude": 11.685674,
            "elevation": 1305.7
        },
        {
            "latitude": 46.474929,
            "longitude": 11.745668,
```

```
                "elevation": 1410.9
            },
            {
                "latitude": 46.474809,
                "longitude": 11.745766,
                "elevation": 1410.4
            },
            {
                "latitude": 46.474929,
                "longitude": 11.745668,
                "elevation": 1410.9
            },


            {
                "latitude": 46.474929,
                "longitude": 11.745668,
                "elevation": 0.0,
                "distance": 13698.295,
                "information": {
                    "name": "water",
                    "coursePointType": "WATER"
                }
            }
        ],
        "activityType": "GRAVEL_CYCLING",
        "coordinateSystem": "WGS84"
    }
```


3.3 Create

This request is to *create* a Course
Method & URL: POST https://apis.garmin.com/training-api/courses/v1/course
Request body: the new course in JSON. A course ID should not be included.
Content-Type: application/json
Response Body: The newly created course as JSON.

Response code:

HTTP Response Status	Description
200	Course creates successfully
401	User access token doesn't exist
412	User permission error
429	Quota violation / rate-limiting

3.4 Retrieve

This request is to *retrieve* a Course
Method & URL: GET https://apis.garmin.com/training-api/courses/v1/course/{courseid}
Response Body: The retrieved course in JSON.

Response code:

HTTP Response Status	Description
200	Course successfully retrieved
401	User access token doesn't exist
412	User permission error
429	Quota violation / rate-limiting

3.5 Update

This request is to *update* a Course
Method & URL: PUT https://apis.garmin.com/training-api/courses/v1/course/{courseid}
Request body: The full updated course in JSON.
Content-Type: application/json
Response Body: The newly created course as JSON.

Response code:

HTTP Response Status	Description
204	Course successfully updated
401	User access token doesn't exist
404	Not Found
412	User permission error
429	Quota violation / rate-limiting

3.6 Delete

This request is to *delete* a Course
Method & URL: DELETE https://apis.garmin.com/training-api/courses/v1/course/{courseid}
Response code:

HTTP Response Status	Description
204	Course successfully updated
401	User access token doesn't exist
412	User permission error
429	Quota violation / rate-limiting

3.7 Acceptance Criteria

- As a best practice, apps should provide course points every 100 meters to align with the Garmin Connect course point and elevation calculations. If an app did not provide it, Garmin Connect will fill in the elevation.
- If the elevation is provided by the app- it will be used. When not provided – Garmin Connect (GC) will use corrected elevation to supplement.
- Apps should provide segments start and end and a valid UUID, to consider as a valid segment in the course path.
- The courses created from 3rd party apps will be visible in GC; these courses will be directly synced to the device. The intent of the API is to fundamentally support individual route syncs. However, up to 50 courses may be synced in a single instance. This limit is set because the courses automatically sync to devices, which can cause undesirable sync times to devices.
- For devices that don't support navigations by default, GC calculates navigations and sends it to the FIT file. We don't provide API access for the apps to send us the navigations (right turn, left turn, etc.)

Limitations with regards to size of a course

- For exporting tcx or gpx files to devices (older devices which are more than 6 years old and only supports tcx and gpx files), there is a limitation of 100 miles for a course.
- We only support embedded turn navigation in the Course FIT file for courses that are 200 miles or less. Some entry-level devices require these turn geo-points to utilize turn-by-turn navigation. Devices with preloaded navigable maps won't need embedded turns to use turn-by-turn navigation.
- Apps can send us around 10,000 geo-points, given any two consecutive geo-points are not more than 100 meters apart, which limits the course to be around 600 miles.